Weak Vector-Valued Hardy Spaces and Finite Dimensionality

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Source And Target

The source paper is F. J. Bertoloto, *Duality of certain Banach spaces of vector-valued holomorphic functions*, arXiv:1203.5322.

For a complex Banach space F and $1 \le p < \infty$, the strong vector-valued Hardy space $H^p(\mathbb{D}; F)$ consists of holomorphic $f : \mathbb{D} \to F$ such that

$$\|f\|_{H^p(F)} = \sup_{0<r<1} \left(\frac{1}{2\pi} \int_0^{2\pi} \|f(re^{it})\|^p \, dt \right)^{1/p} < \infty.$$

The weak space is

$$H^p(\mathbb{D}; F)_w = \{f \in \mathcal{O}(\mathbb{D}; F) : \ x^* \circ f \in H^p(\mathbb{D}) \text{ for every } x^* \in F^*\},$$

with norm

$$\|f\|_{w,p} = \sup_{\|x^*\| \le 1} \|x^* \circ f\|_{H^p}.$$

The source ends with Conjecture 5.11:

Conjecture 5.11. We conjecture that $H^p(\mathbb{D}; F) = H^p(\mathbb{D}; F)_w$ if and only if F is finite dimensional.

The statement is intended for $1 \le p < \infty$: the source's Proposition 5.9 already records that $H^\infty(\mathbb{D}; F)_w = H^\infty(\mathbb{D}; F)$ for every Banach space F.

Result

Theorem 1. *Let $1 \le p < \infty$ and let F be a complex Banach space. Then*

$$H^p(\mathbb{D}; F) = H^p(\mathbb{D}; F)_w$$

as sets if and only if F is finite-dimensional.

Proof Intuition

The weak Hardy norm only tests scalar images of a vector-valued holomorphic function. In an infinite-dimensional Banach space, Dvoretzky-Rogers gives a sequence (x_n) that looks ℓ^p after every scalar functional, but whose norms are not ℓ^p-summable. Hardy interpolation lets us build a holomorphic function f taking the normalized interpolation values (x_n). Scalar interpolation makes $x^* \circ f \in H^p$ for every x^*, while the vector-valued Carleson embedding theorem says that any strong $H^p(\mathbb{D}; F)$ function must have ℓ^p-summable normalized values on that same interpolating sequence. Thus f is weak Hardy but not strong Hardy.

Standard Inputs

Standard Fact 1 (Dvoretzky-Rogers)**.** *For $1 \le p < \infty$, the identity map on a Banach space F is absolutely p-summing if and only if F is finite-dimensional. Equivalently, if F is infinite-dimensional, there is a sequence $(x_n) \subset F$ such that*

$$\sup_{\|x^*\| \le 1} \sum_{n=1}^{\infty} |x^*(x_n)|^p < \infty$$

but

$$\sum_{n=1}^{\infty} \|x_n\|^p = \infty.$$

Standard Fact 2 (Hardy interpolation and Carleson evaluation)**.** *For every $1 \le p < \infty$ there are points $a_n \in \mathbb{D}$ and functions $h_n \in H^p(\mathbb{D})$ such that:*

1. *$(1 - |a_m|^2)^{1/p} h_n(a_m) = \delta_{mn}$;*
2. *for every scalar sequence $c = (c_n) \in \ell^p$, the series $\sum_n c_n h_n$ converges in H^p, locally uniformly on $\mathbb{D}$, and satisfies*
$$\left\| \sum_n c_n h_n \right\|_{H^p} \le C_p \|(c_n)\|_{\ell^p};$$
3. *the functions may be chosen locally summable, $\sum_n |h_n(z)| < \infty$ locally uniformly in $z \in \mathbb{D}$;*
4. *for every Banach space F and every $G \in H^p(\mathbb{D}; F)$,*
$$\sum_{n=1}^{\infty} (1 - |a_n|^2) \, \|G(a_n)\|^p \le C'_p \|G\|^p_{H^p(F)}.$$

This is the standard interpolation theorem for H^p on an interpolating Blaschke sequence, together with the Carleson embedding theorem. The last inequality is the vector-valued form and follows immediately from applying the scalar Carleson measure estimate to the subharmonic function $\|G(\cdot)\|^p$.

Proof

First suppose F is finite-dimensional. Let $e_1, \ldots, e_d$ be a basis of F, with dual coordinate functionals $e_1^*, \ldots, e_d^*$. If $f \in H^p(\mathbb{D}; F)_w$, then $e_j^* \circ f \in H^p(\mathbb{D})$ for every j, and

$$f = \sum_{j=1}^{d} (e_j^* \circ f) e_j.$$

Since all norms on F are equivalent, this finite coordinate decomposition gives $f \in H^p(\mathbb{D}; F)$. The reverse inclusion is immediate from $|x^*(f)| \leq \|x^*\| \|f\|$. Thus equality holds when F is finite-dimensional.

Now suppose F is infinite-dimensional. Choose $(x_n) \subset F$ as in the Dvoretzky-Rogers fact. Let a_n, h_n be the Hardy interpolation data above, and define

$$f(z) = \sum_{n=1}^{\infty} h_n(z) x_n, \qquad z \in \mathbb{D}.$$

The local summability of (h_n) and boundedness of every weakly ℓ^p-summable sequence imply that this series converges locally uniformly in F, so $f \in \mathcal{O}(\mathbb{D}; F)$.

For $x^* \in F^*$, the scalar function is

$$x^* \circ f = \sum_{n=1}^{\infty} x^*(x_n) h_n.$$

Since $(x^*(x_n)) \in \ell^p$, the interpolation estimate gives

$$\|x^* \circ f\|_{H^p} \leq C_p \left(\sum_{n=1}^{\infty} |x^*(x_n)|^p \right)^{1/p}.$$

Taking the supremum over $\|x^*\| \leq 1$ shows $f \in H^p(\mathbb{D}; F)_w$.

On the other hand, the interpolation normalization gives

$$(1 - |a_n|^2)^{1/p} f(a_n) = x_n, \qquad n \geq 1.$$

If f belonged to $H^p(\mathbb{D}; F)$, then the vector-valued Carleson evaluation estimate would yield

$$\sum_{n=1}^{\infty} \|x_n\|^p = \sum_{n=1}^{\infty} (1 - |a_n|^2) \|f(a_n)\|^p \leq C_p' \|f\|_{H^p(F)}^p < \infty,$$

contradicting the choice of (x_n). Hence

$$f \in H^p(\mathbb{D}; F)_w \setminus H^p(\mathbb{D}; F).$$

So equality fails for every infinite-dimensional F, and the theorem is proved. $\square$

Verification And Novelty Check

The proof uses only standard Hardy-space interpolation, the Carleson embedding theorem, and the Dvoretzky-Rogers theorem on absolutely p-summing identity operators. The $p = \infty$ endpoint is excluded because the source itself proves equality there for every F.

The run's cheap indexes had no prior packet for arXiv:1203.5322. A bounded web search on 2026-07-03 for the exact phrase

$$H^p(\mathbb{D}; F) = H^p(\mathbb{D}; F)_w$$

and variants involving "weak Hardy space", "finite dimensional", and "Dvoretzky-Rogers" found the source arXiv page and no later explicit answer.

Human Review Recommendation

Review as a likely full solution to Conjecture 5.11 for $1 \leq p < \infty$. The main item to check is the stated Hardy interpolation package: one needs an interpolating sequence whose interpolation functions are locally summable, so that $\sum h_n x_n$ is an honest F-valued holomorphic function, and whose normalized point evaluations satisfy the vector-valued Carleson estimate.

References

References

[1] F. J. Bertoloto, *Duality of certain Banach spaces of vector-valued holomorphic functions*, arXiv:1203.5322.

[2] P. L. Duren, *Theory of H^p Spaces*, Academic Press, 1970.

[3] J. B. Garnett, *Bounded Analytic Functions*, Academic Press, 1981.

[4] A. Pietsch, *Operator Ideals*, North-Holland, 1980.